ADR-0001: Data source strategy for calibration and validation

Status: accepted Date: 2026-07-23 (accepted 2026-07-23 — Will had no strong preference and deferred to this recommendation)

Context

The model needs two things from historical Jeopardy! data: (1) aggregate statistics to calibrate the win-probability engine (DD accuracy by round and row, human FJ wagering distributions by rank), and (2) a way to validate that the engine's predictions match real outcomes.

J-Archive's site terms forbid scraping, republishing, or monetizing its content. The most complete derived dataset available, jwolle1's `jeopardy_clue_dataset` on GitHub (538,845 clues as of the Season 41 release; the README listed 554,131 clues through Season 42 in July 2026; with Daily Double wager amounts, contestant scores, Coryat scores and right/wrong counts), carries its own restriction: its README says not to use the data "to make a public-facing web site, app, or any other product," and the source prefers not to be credited.

Tesauro et al. (2013), published in the Journal of Artificial Intelligence Research, mined roughly 3,000 J-Archive episodes and published the resulting aggregate statistics (DD accuracy by row, human wager distributions, etc.) as part of a peer-reviewed paper. These are citable published findings, not raw scraped data, and carry no reuse restriction.

Decision

- Do not scrape J-Archive directly.
- Use the Tesauro et al. published statistics as the calibration source for anything that ships: the white paper, the deck, the technical docs, and any numbers quoted in the tool's output.
- Use the jwolle1 dataset only inside the internal validation harness (tests, simulator calibration checks) and never surface raw rows, aggregate tables derived directly from it, or the dataset itself in a shipped artifact.
- If a shipped artifact needs a statistic the paper does not cover, generate it from our own simulation rather than from the restricted dataset, and label it clearly as simulated, not historical.

Consequences

- The white paper's data appendix will cite one peer-reviewed source instead of a larger raw dataset, which narrows what we can claim with historical backing. Flag any claim that isn't in the paper as "simulated" rather than "observed."

- Internal validation can still be strong, since the jwolle1 dataset remains usable for testing and calibration — it just cannot appear in outputs.
- If Will wants richer historical detail in the shipped product later, this ADR should be revisited — for example, by seeking direct permission from a data holder, or by running our own compliant, permitted data collection.